

Algorithmic Redistricting: A Non-Partisan Solution for State Legislatures

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Executive Summary

The Problem. Politicians draw their own district lines. This lets them choose their voters before voters choose them. Ninety percent of seats are effectively uncontested. Minority communities are underrepresented. Trust in elections falls.

The Solution. Computer algorithms draw districts from census geography alone. They see no voter registration data and no election results. They cannot be instructed to favor any party.

The Evidence. Research across all 50 states and three census decades shows: **20% more compact districts, 69 more majority-minority districts, and 62% less partisan bias** compared to enacted plans.

The Path Forward. Three adoption models fit different state contexts: legislative mandate, commission guidelines, or court-ordered remedy. Estimated cost: \$50K–\$200K versus \$1M+ for redistricting litigation.

The Ask. Consider algorithmic redistricting for your state’s next cycle. The 2030 redistricting begins in 2031. States that act now will have time to implement and build public confidence.

The Problem

Politicians draw their own districts. This is a conflict of interest with measurable consequences.

Every ten years, states redraw congressional and legislative district lines after the census. In most states, the legislature draws those maps. Elected officials design the districts they will run in.

The result is gerrymandering—manipulating boundaries for partisan gain. Mapmakers use two techniques. They *pack* opposition voters into a few districts, wasting their votes. They *crack* the rest across many districts, diluting their influence.

The consequences are measurable:

- **90% of seats** are safe. General elections rarely determine outcomes.
- **Voter trust falls.** Most Americans believe their district is predetermined.
- **Minorities lose seats.** Drawn maps produce 69 fewer majority-minority districts than algorithmic alternatives.
- **Bizarre shapes.** Districts span hundreds of miles to connect voters with no shared interests.

Litigation is the current remedy. It costs \$1M to \$5M per state. Cases take years. By the time courts act, another election has passed under the flawed map.

This problem has a ready, tested, and affordable solution.

The Solution

The algorithm draws districts from geography and population. It has no access to political data.

Algorithmic redistricting uses open-source software to draw district lines automatically. The algorithm starts with census tracts—small geographic units with known populations. It builds a map of which tracts share a border. Then it splits the state into equal-population districts. At each split, it minimizes the length of the dividing line. The result: compact districts that follow natural geographic boundaries.

Think of it like a GPS for redistricting. You provide the destination (number of districts and population targets). The algorithm finds the most geographically natural route. No political detours are possible.

What goes in: census population counts, geographic boundaries, number of districts required by law.

What stays out: voter registration, past election results, race, party, and incumbency data.

The structural guarantee. The algorithm cannot gerrymander because it cannot *see* partisan data. This is not a policy choice. It is a structural property. The same inputs always produce the same map. Anyone can verify this in minutes using open-source code.

The algorithm runs in minutes, not months. The code is publicly available. Any university, court, or journalist can audit the process independently.

Six Key Findings

Algorithmic maps outperform enacted maps on every measurable dimension of fairness.

Research covers all 50 states and three census decades (2000, 2010, 2020). All findings compare algorithmic maps to current enacted plans.

Finding 1: 20% More Compact

B.1, B.2

Algorithmic districts are 20% more compact than enacted maps. Illinois improves by 174%. Maryland improves by 99%. North Carolina improves by 76%. Compact districts respect natural community boundaries. They are far harder to manipulate than the winding shapes drawn for political advantage.

Finding 2: 69 More Majority-Minority Districts *D.0*

Algorithmic maps create 69 more majority-minority districts nationally. The algorithm uses no racial data. Yet geographic concentration of minority communities means compact, equal-population districts naturally capture minority representation. The algorithm exceeds Voting Rights Act requirements without explicit racial targeting.

Finding 3: 62% Less Partisan Bias

C.5

Partisan bias—measured by the efficiency gap, a standard legal metric—drops from +5.1% to −3.2% under algorithmic maps. That is a 62% reduction. Neither party gains a structural advantage from the map itself. Both parties compete on a level geographic playing field.

Finding 4: 30% More Competitive Districts

C.8

The share of genuinely competitive districts rises by 30%. Candidates must appeal to a broader electorate. General elections become meaningful again. Winning requires persuasion, not just mobilization of a pre-sorted base.

Finding 5: Stable Across Decades

B.15, C.3

The same geographic logic applied to 2000, 2010, and 2020 census data produces similar district shapes each decade. Boundaries remain stable. Communities stay in their districts. Voters know their representative. Political disruption from redistricting

decreases substantially.

Finding 6: Works for State Chambers Too

F.1

The algorithm scales to any district count. It has been tested on all 50 state houses—from Delaware’s 41-seat chamber to New Hampshire’s 400-seat chamber. It handles dense urban states and sprawling rural ones. It accommodates islands, water crossings, and irregular borders.

Three Ways to Adopt

States can adopt algorithmic redistricting through legislation, commission guidelines, or court remedy.

Model A: Legislative Mandate

The legislature passes a law requiring algorithmic redistricting. The law specifies algorithm parameters, population tolerance, and an independent audit process. Any state with appetite for redistricting reform can pursue this model.

- **Timeline:** 1–2 legislative sessions
- **Cost:** \$50K–\$100K (software, training, audits)
- **See:** Appendix A for model statutory language

Model B: Commission Guidelines

An independent redistricting commission adopts the algorithm as a neutral baseline. The commission retains authority to adjust for communities of interest. The algorithm provides a starting point. The commission ensures legal compliance and public accountability.

- **Applies to:** 14 states with independent commissions
- **Timeline:** 6–12 months before the next cycle
- **Cost:** \$75K–\$150K (software, training, public engagement)

Model C: Court-Ordered Remedy

Courts can order algorithmic redistricting when striking down a gerrymandered map. The algorithm provides a neutral, reproducible baseline for judicial review. Courts in Pennsylvania and North Carolina have used similar approaches.

- **Applies to:** Any state with active redistricting litigation
- **Timeline:** Deployable immediately as emergency remedy
- **Cost:** \$100K–\$200K (expert witnesses, implementation)

Implementation Checklist

- ☐ Legal review: confirm state constitution permits algorithmic approach
- ☐ Software: open-source code is available at no cost
- ☐ Data: census redistricting files are publicly available
- ☐ Capacity: hire a GIS analyst or partner with a state university
- ☐ Validation: commission an independent audit of outputs
- ☐ Engagement: hold hearings and publish interactive maps online
- ☐ Start at least two years before the redistricting cycle

Addressing Common Concerns

Every major objection has a direct, evidence-based answer.

Q: Can this produce gerrymandered maps?

A: No. The algorithm has no access to voter registration data or election results. It cannot favor any party. This is not a policy choice. It is a structural property of how the algorithm works.

Q: What about minority voting rights?

A: The algorithm produces 69 *more* majority-minority districts than current maps. It meets Voting Rights Act requirements without using racial data. Geographic concentration of minority communities naturally produces representative districts under compact, equal-population rules.

Q: Will this eliminate competitive districts?

A: The opposite is true. Algorithmic maps produce 30% *more* competitive districts than current enacted plans. Gerrymandered maps eliminate competition by design. Algorithmic maps restore it by following geography.

Q: Does this work for our state house?

A: Yes. The algorithm scales to any district count. It has been tested on every state house—from Delaware’s 41 seats to New Hampshire’s 400 seats. It performs consistently across urban, rural, and mixed geographies.

Q: How much does this cost?

A: Implementation runs \$50K–\$200K depending on adoption model. Redistricting litigation typically costs \$1M to \$5M per state—and often produces another round of litigation. The census data is free. The software is open source.

Q: Who else supports this approach?

A: Public legitimacy research shows broad bipartisan support when voters learn that the algorithm cannot access party data (C.6). Transparency builds trust. The open-source code means any group can audit the process. No single party or entity controls the outcome.

No catch. The only parties who face greater competition are those who currently benefit from drawn district lines.

Call to Action

Three concrete steps for legislative staff.

Step 1: Request a state demonstration. State-specific algorithmic maps are available for all 50 states across three census years. See what the districts look like before committing to any policy.

Step 2: Introduce model legislation. Appendix A contains draft statutory language. Legislative staff can adapt it to your state’s constitutional requirements. Technical assistance is available at no cost.

Step 3: Download and verify the code. The algorithm is open source. Your state university can run it independently. Reproducibility is a core feature—not a claim. Anyone can confirm that the same inputs produce the same maps.

Algorithmic redistricting is practical, proven, and ready for adoption. States that act now will have time to implement, audit, and build public confidence before the 2031 redistricting cycle.

Appendix A: Model Statutory Language

Draft text adapted from research on algorithmic redistricting implementation (D.4). States should obtain independent legal review before introduction. Bracketed items require state-specific completion.

AN ACT relating to congressional redistricting; requiring the use of algorithmic redistricting methods; providing definitions; establishing validation requirements.

Section 1 — Findings.

The Legislature finds that:

- (a) Current redistricting processes create conflicts of interest when elected officials draw the districts they will run in.
- (b) Algorithmic redistricting provides an objective, transparent, and reproducible alternative that uses only population and geographic data.
- (c) Research across all 50 states demonstrates that algorithmic methods produce districts that are more compact, create more majority-minority districts, and reduce partisan bias compared to legislatively drawn maps.
- (d) Algorithmic redistricting is cost-effective: implementation costs of \$50,000–\$200,000 compare favorably to redistricting litigation costs of \$1,000,000 or more.

Section 2 — Definitions.

As used in this Act:

- (a) “*Algorithmic redistricting*” means a computer-based method of drawing district boundaries using only census population data and geographic boundary data, with no access to voter registration, party affiliation, or election results.
- (b) “*Compactness*” means geometric compactness measured by the Polsby-Popper score ($4\pi \times \text{Area}/\text{Perimeter}^2$), where 1.0 represents a perfect circle.
- (c) “*Baseline map*” means the initial district map produced by the algorithm before any adjustments.
- (d) “*Independent audit*” means a review conducted by a qualified party with no interest in the redistricting outcome.

Section 3 — Requirements.

redistricting authority shall use algorithmic redistricting to produce a baseline map for [congressional / state legislative] districts following each decennial census.

- (a) The algorithm shall use the following parameters: population balance tolerance of no more than 0.5%; geographic contiguity required for all districts; compactness optimization enabled; no input of voter registration, party, or election data.
- (b) The baseline map shall be published for public comment no fewer than 60 days before adoption of any final plan.

Section 4 — Validation.

- (a) An independent audit shall be conducted by [a state university or qualified nonpartisan research organization] before publication of the baseline map.
- (b) The audit shall confirm: (i) population equality within the required tolerance; (ii) geographic contiguity of all districts; (iii) compliance with the federal Voting Rights Act; (iv) reproducibility—that running the algorithm with the same inputs produces the same map.
- (c) The audit report shall be published online before any legislative vote on a final plan.

Section 5 — Adjustments.

- (a) The redistricting authority may adjust the baseline map to preserve communities of interest, comply with the Voting Rights Act, or address geographic anomalies.
- (b) Any adjustment shall: (i) maintain population balance within the required tolerance; (ii) not reduce the compactness score of any affected district by more than 5% from the baseline; (iii) be accompanied by written justification describing the community of interest or legal requirement addressed.
- (c) Adjustments may not be made for the purpose of favoring or disfavoring any political party, incumbent, or candidate.

Section 6 — Implementation.

- (a) This Act takes effect [date at least two years before the next redistricting cycle].
- (b) There is appropriated from the General Fund the sum of \$[amount] for software procurement or development, staff training, independent audit, and public engagement.
- (c) The redistricting authority shall publish all algorithm source code, input data, and output maps on a public website within 30 days of producing any baseline map.

Appendix B: Cost Comparison

Estimated costs for redistricting approaches in a typical state. Litigation figures are based on publicly reported cases in Pennsylvania, North Carolina, Maryland, and Ohio.

Approach	Low Estimate	High Estimate	Notes
<i>Algorithmic Redistricting</i>			
Legislative mandate	\$50,000	\$100,000	Software, training, independent audit
Commission adoption	\$75,000	\$150,000	Software, training, public engagement
Court-ordered remedy	\$100,000	\$200,000	Expert witnesses, technical implementation
<i>Traditional Redistricting</i>			
Legislative process	\$200,000	\$500,000	Staff, consultants, hearings
Redistricting litigation	\$1,000,000	\$5,000,000	Legal fees, expert witnesses
Federal court remedy	\$500,000	\$2,000,000	Court-appointed expert, legal fees
<i>Recurring Costs (each redistricting cycle)</i>			
Algorithmic re-run	\$10,000	\$25,000	New census data, updated audit
Legislative re-draw	\$200,000	\$500,000	Same process repeated every 10 years
Litigation risk	variable	variable	If current enacted map is challenged

Cost Component	Notes
Census data	Publicly available at no cost from the U.S. Census Bureau
Algorithm software	Open-source code available at no cost; commercial support optional
GIS analyst	One FTE for 6–12 months (\$60K–\$100K loaded cost)
Independent audit	\$20K–\$50K; university partnership may reduce cost
Public engagement	\$15K–\$30K for hearings, website, and interactive maps
Legal review	\$15K–\$40K for constitutional and VRA compliance review

Bottom line. A state adopting algorithmic redistricting through a legislative mandate can expect to spend \$50K

to \$100K total. That is less than the legal fees for a single redistricting lawsuit. States with ongoing redistricting litigation may recoup implementation costs within the first year by reducing or eliminating active legal proceedings.